

Music-Gen v4 — Cycles 53-55

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Music-Gen v4 — Cycles 53-55

Abstract

Cycles 53-55 extended the preservation-only heartbeat cadence into its twelfth, thirteenth, and fourteenth consecutive substantive-heartbeat cycles while all six operator-authority escalations remained preserved verbatim with `blocked_on_operator=true`. The range formalized two new selection-event patterns for surfaces previously tracked only in cycle narrative — a canonical Priority 0 escalation-preservation sidecar (established at Cycle 53, chain-superseded across Cycles 54 and 55) and a canonical Priority 5 Track B/C/D deferral-preservation rollout sidecar (established at Cycle 53) — and adopted a HOLD-pattern selection event for the consolidation-proposal question opened in the prior range’s Cycle 50 (established at Cycle 54 with `_selection/c44-consolidation-proposal-hold.json` SHA `f70b8ab1...`; chain-superseded across Cycle 55 with `c45-consolidation-proposal-hold.json`). Cycle 55 also surfaced a first-class positive downstream signal: a fresh filesystem probe found `docs/specs/v4_sound_matching_layer_spec.md` PRESENT on disk (previously ABSENT across Cycles 53 and 54); per FD-1 + invariant (d) the worker correctly disclosed the change rather than attempting to author the doc, and the audit endorsed the closure as I-5-closed for downstream M-V4-PROFILES scoping. Cycle 55 also adopted the c45 brief’s I-1 narrative-counter realignment across all six escalation memos, bumping `carried_from_cycle` from `{15, 15, 15, 14, 14, 13}` at Cycle 54 close to `{16, 16, 16, 15, 15, 14}` at Cycle 55 close so the narrative counters and the on-disk chain agree. The Priority 1 chain-supersede lineage on the workspace-forced OPT_B ledger-emitter exemption now runs eleven consecutive cycles past the c34 fork; the Priority 2 stand-pat lineage on the POR shadow-drift strengthening question now runs twelve consecutive cycles past the c34 empirical proof. All six escalation sidecars under `data/v4/_manager/` verified byte-identical pre-vs-post across the range with the standard six-row inventory (non-CG bass, metric-semantics, drums-fine, v2-bass-fine, guitar-fine, composite-FP-drift); `_manager/` exhaustive at six; SF2_CONFIRMED remains FORBIDDEN on non-CG bass. Test coverage grew in-place from `34 → 36 → 38` cases on `tests/test_c30_legacy_mode_regression.py` across the range with the standalone eight-case OP-1 serial-lock suite unchanged; cross-cycle total 48/48 PASS at range close (test_39 pinning the c14 string-supersedes_path lemma across all six c45 sidecars, test_40 pinning the P0 sidecar shape and the before-vs-after invariant on the six escalation

memos). POR parseable-Milestones traced 873 → 887 → 901 → 915 across the range in a stable +14/cycle rhythm (structurally attributable per the c34 empirical proof to four Track B/C/D deferral rows + housekeeping tail + preservation events + test-extension rows plus the standing Priority 8 preservation-and-scan row); each cycle's open matched the immediately-prior cycle's close exactly. Nine read-only anchors byte-identical pre-vs-post at range close including the c40 consolidation strategy proposal. Cycle 55 also cosmetically surfaced ten pre-existing ledger rows tagged cycle=45 at lines 727-736 originating from an earlier unrelated M-EAR-1 real-label-training arc; no event_id UUID5 content-hash collision; the shared milestone_id labels (_run/cycle_45_closed, _archive/cycle-45-scratch, _infra/adopt-cycle45-tests) are a v3→v4 campaign-continuity naming artifact, not a defect. Independent audit returned **VALIDATED** across all nine validators (a)-(i) with four informational disclosures (I-1 the spec doc now PRESENT and endorsed as I-5-closed; I-2 the worker-side D-3 misreading pattern recurring across five cycles and worth an explicit brief flag; I-3 the cosmetic v3-era vs v4-era cycle-45 naming collision; I-4 the standing closure-vs-preservation scope divergence carried across fourteen consecutive cycles). Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Canonical 7-key env_pin_sha256=2ac444c3...922ca unchanged; FD-16(a) re-issue not triggered. M-V4-SHOWCASE-1 status unchanged (LANDS_pending_operator; the CG A/B mix WAV holds SHA 6e13e007... byte-identical since c17).

Introduction

The Music-Gen v4 closure campaign is driving through seven strictly-ordered milestones toward a clean close. Cycles 53-55 are the twelfth, thirteenth, and fourteenth consecutive substantive-heartbeat cycles under the c36 auditor's terminal contracts. All substantive stage-2 sweep work remains gated on operator adjudication that has not arrived through live_guidance. The prior range added a fourth operator-authority surface — the POR consolidation strategy proposal opened at Cycle 50, awaiting operator selection among three named options (OPT_1 aggressive compaction, OPT_2 progressive compaction, OPT_3 retain current pattern) — and the current range formalizes two additional selection-event patterns for surfaces that were previously tracked only in cycle narrative.

The range also surfaces one genuinely new positive downstream signal: a spec document (docs/specs/v4_sound_matching_layer_spec.md) that had been ABSENT across the two prior cycles is now PRESENT on disk at Cycle 55. Per FD-1 the worker correctly disclosed the change rather than attempting to author the doc — authoring would have violated the FD-1 authoring boundary — and the audit endorsed the closure as I-5-closed. This is not a state change the campaign engineered; it is a state change the campaign detected honestly and forwarded to the next research pass as a positive scoping signal for M-V4-PROFILES readiness.

Approach

Each cycle executed a nine-priority template, unchanged in shape from the prior range but extended in preservation-event coverage across the range:

- **Priority 0 (preserve six escalations).** Verify all six escalation sidecars under data/v4/_manager/ byte-identical versus the prior cycle's attestation SHAs. Emit canonical P0 escalation-preservation sidecar chain-superseding the prior cycle's P0 sidecar via string supersedes_path per the c14 lemma (established at Cycle 53; the pattern makes the P0 preservation itself an on-disk artifact rather than only a narrative claim, giving audit surface for the before-vs-after invariant across the six memos). Bump carried_from_cycle per calibration.
- **Priority 1 (chain-continuation re-probe).** Execute test -d long_exposure, confirm ABSENT, emit chain-supersede preservation event with string supersedes_path per the c14

lemma pointing at the prior cycle’s actual on-disk filename per the M-1 canonical naming.

- **Priority 2 (POR stand-pat continuation).** Scan `live_guidance` for operator-supplied c31/c32 POR snapshot, confirm absent, emit stand-pat preservation event chain-superseding the prior cycle’s stand-pat via string `supersedes_path`.
- **Priority 3 (composite-FP-drift operator adjudication).** Scan for `PATH_A` / `PATH_B` / `PATH_C` adjudication token, confirm absent, correctly skip.
- **Priority 4 (Peach Dream stem manifest anchor).** Verify SHA c4944ee80dfe446b... byte-identical; carry invariant-(d) divergence disclosure on the `operator_section_c25_checkpointed/rc9_6stem/` non-standard path on the deferral row; tenth consecutive preservation.
- **Priority 5 (honest-deferral rows for Track B/C/D).** Emit four rows with concrete resume commands + one canonical rollup preservation supersede (established at Cycle 53; the pattern makes the Track B/C/D preservation itself an on-disk artifact chain-superseding across cycles).
- **Priority 6 (POR shadow-zone hold verification).** Verify parseable_milestones at cycle open matches prior cycle close (delta zero); emit fresh-hold event with `supersedes_path`: null per c14 lemma; post-registration count grows by exactly the number of ledger rows emitted this cycle.
- **Priority 7 (Track F test-suite extension).** Extend `tests/test_c30_legacy_mode_regression.py` in-place by two cases per cycle capturing the cycle’s new preservation events’ shape and chain-integrity.
- **Priority 8 (consolidation-proposal preservation-and-scan / HOLD).** At Cycle 54, established the canonical HOLD pattern for the consolidation-proposal question — verify the c40 proposal doc SHA byte-identical (8cffclcecf8fed87...), scan `live_guidance` for `OPT_1` / `OPT_2` / `OPT_3` selection token, emit `_selection/cN-consolidation-proposal-hold.json` when absent. At Cycle 55, chain-superseded the c44 hold via string `supersedes_path`.

Cycle 53 specific. Established the canonical Priority 0 escalation-preservation sidecar pattern and the canonical Priority 5 Track B/C/D deferral-preservation rollup sidecar pattern.

Cycle 54 specific. Established the canonical Priority 8 consolidation-proposal HOLD sidecar pattern (`_selection/c44-consolidation-proposal-hold.json` SHA f70b8ab1...); chained the P0 sidecar to c44 (608e8138...); chained the P1 preservation to c44 (53faa9a2...); chained the P2 stand-pat to c44 (98e01b20...).

Cycle 55 specific. Adopted the c45 brief’s I-1 narrative-counter realignment across all six escalation memos, bumping `carried_from_cycle` values from {15, 15, 15, 14, 14, 13} at c44 close to {16, 16, 16, 15, 15, 14} at c45 close so the brief and on-disk chains agree. Discovered `docs/specs/v4_sound_matching_layer_spec.md` PRESENT on disk (previously ABSENT across c43/c44), disclosed honestly per FD-1 + invariant (d) via `c45-consolidation-proposal-hold.json.i5_binding_doc_absent_check`, did not attempt authoring. Cosmetically surfaced ten pre-existing ledger rows tagged `cycle=45` at lines 727-736 from an earlier M-EAR-1 real-label-training v2 arc — no `event_id` UUID5 content-hash collision; different content, same milestone_id label class — as a v3→v4 campaign-continuity naming artifact rather than a defect. Landed the standard six sidecars (P0 escalation-preservation, P1 emitter-writer-boundary, P2 stand-pat, P5 Track B/C/D deferral rollup, P8 consolidation-proposal hold, plus one fresh-hold P6 with `supersedes_path`: null).

Discipline guards asserted across the range. All AST-scannable invariants pass: no PRNG imports (`random.*`, `np.random.*`), no `sidecar_nonfactor`, no VST3 state APIs, no `--verify-det` by-pass, `/usr/bin/python3` interpreter guard implicit via the c34+ pattern. `supersedes_path` typed as string or null throughout the range per c14 lemma across all c45 sidecars alone (six string, one

null). Canonical 7-key env_pin_sha256=2ac444c36298d6ada0579abala9160a5881703a4e628f5ccdd828b842a922ca unchanged; FD-16(a) re-issue not triggered. FD-16(c) replay-proof invariant carried; no new render surfaces. OP-1 serial-lock invariant remains in force; the 8/8 standalone lock tests continue to guard. Zero SF2_CONFIRMED verdicts on non-CG bass. No wait-on-operator memo (banned per operator directive 2026-09-03 part 2); Priority 0 preserve-only posture satisfies the genuine operator-authority carve-out.

Findings

Two new canonical selection-event patterns established

Two priorities that had previously been tracked only through cycle narrative received canonical on-disk selection-event patterns during the range:

- **Priority 0 escalation-preservation sidecar** (established Cycle 53). `_selection/cN-escalation-preservation.json` chain-supersedes the prior cycle's P0 sidecar via string `supersedes_path` and pins the six escalation memos' before-vs-after SHAs. This makes the P0 preservation itself an on-disk artifact rather than only a narrative claim, giving audit surface for the before-vs-after invariant. Cycle 55's c45 sidecar chain-supersedes the c44 sha 608e8138... per I-2 canonical adoption.
- **Priority 5 Track B/C/D deferral-preservation rollup sidecar** (established Cycle 53). `_selection/cN-track-bcd-deferral-preservation.json` chain-supersedes the prior cycle's P5 rollup and carries the four honest-deferral rows as a stable summary rather than as scattered narrative. Cycle 55's c45 sidecar sha db8bfed7....

Both patterns exercise the c14 string-typed `supersedes_path` lemma per cycle and are on-disk discoverable via `ls data/v4/_selection/`.

Consolidation-proposal HOLD pattern established at Cycle 54

`_selection/c44-consolidation-proposal-hold.json` (SHA f70b8ab1...) formalized the Priority 8 preservation-and-scan discipline as an on-disk artifact. The pattern: verify the c40 proposal doc SHA byte-identical (8cffc1cecf8fed87...), scan `live_guidance` for `OPT_1 / OPT_2 / OPT_3` selection token, emit HOLD sidecar when absent. Cycle 55's `c45-consolidation-proposal-hold.json` chain-supersedes c44 via string `supersedes_path`.

Positive downstream signal — spec doc now PRESENT on disk

Cycle 55's I-5 finding: a fresh filesystem probe found `docs/specs/v4_sound_matching_layer_spec.md` PRESENT on disk. The doc had been ABSENT across Cycles 53 and 54 and been forwarded as a downstream concern in both briefs for M-V4-PROFILES substantive advance. The Cycle 55 worker performed the fresh probe honestly, disclosed the change per FD-1 + invariant (d), and correctly did not attempt authoring — authoring would have violated the FD-1 authoring boundary. The audit endorsed the closure as I-5-closed and flagged it as a positive downstream signal for the next cycle's M-V4-PROFILES readiness scoping. The state change was not engineered by the campaign; it was detected honestly and forwarded.

Six operator escalations preserved verbatim with I-1 counter realignment

All six escalation sidecars under `data/v4/_manager/` verified byte-identical at Cycle 55 close with `carried_from_cycle` realigned per the c45 brief's I-1:

Escalation	Origin	c44 close	c45 close (I-1 adopted)
SHOWCASE-1-non-cg-bass-acceptance-policy	c7	15	16
M-V4-METRIC-SEMANTICS-c16	c16	15	16
CERT-fine-fit-sf2-drums-legacy-halt	c30	15	16
CERT-fine-fit-sf2-v2-legacy-halt	c31	14	15
CERT-fine-fit-sf2-guitar-legacy-halt	c31	14	15
CERT-composite-fp-drift-adjudication-c32	c32	13	14

The realignment ensures the brief-cited narrative counters match the on-disk chain. All six carry correct `carried_from_cycle` semantics; all `blocked_on_operator=true`; all discoverable via both `grep _manager/M-V4-CERT` and `ls data/v4/_manager/. _manager/` remains exhaustive at six; no seventh escalation opened. `SF2_CONFIRMED` remains FORBIDDEN on non-CG bass.

Chain-supersede lineages matured to eleven and twelve cycles

Priority 1 (workspace-forced `OPT_B` ledger-emitter exemption): eleven consecutive chain-supersede cycles past the c34 fork — c34 fork → ... → c45 (Cycle 55's `c45-emitter-writer-boundary-preservation.json` string-supersedes c44 sha 53faa9a2...).

Priority 2 (POR shadow-drift stand-pat): twelve consecutive stand-pat cycles past the c34 empirical proof — c34 empirical proof → c35 blocker → ... → c45 (Cycle 55's `c45-por-drift-preservation.json` string-supersedes c44 stand-pat sha 98e01b20...). c34 empirical attribution finding + c35 blocker preserved transitively byte-identical.

POR arithmetic +14/cycle across the range

Parseable-Milestones trajectory: 873 (post-c42) → 887 (post-c53) → 901 (post-c54) → 915 (post-c55). Each cycle's post-registration count matches the prior cycle's close-baseline plus exactly 14 ledger rows emitted this cycle. Each cycle's open matches the immediately-prior cycle's close (delta zero). POR baseline 745 (c31 counting-method drift baseline) maintained throughout.

Test coverage grew 34 → 40 in-place

Cross-cycle regression coverage on `tests/test_c30_legacy_mode_regression.py` extended in-place by two cases per cycle across the range: 34 → 36 → 38 → 40. Cycle 55's additions: `test_39` pinning the c14 string-supersedes_path lemma across all six c45 sidecars (verifying no list-typed supersede leaked into the six preservation surfaces); `test_40` pinning the P0 sidecar shape assertion + before-vs-after invariant on the six escalation memos. Standalone `tests/test_fine_fit_serial_lock_c32.py` unchanged at 8/8. Cross-cycle total at range close: 48/48 PASS.

Read-only anchors held; four informational disclosures at Cycle 55 audit

Nine read-only anchors verified byte-identical pre-vs-post at range close: `scripts/sound_match/objective.py` 8087ce80...; `scripts/sound_match/_sweep_hygiene_c27.py` 771ff42b...; `scripts/sound_match/_serial_lock_op1.py` 121809db...; `data/v4/deliveries/31a164f845f8e27e/cg_ab_mix.wav` 6e13e007...; `docs/agent_picks_selection_invariants.md` 29a1610b...; `docs/emitter_exemption_policy.md` fd2c33a7...; `docs/v4_por_consolidation_strategy_proposal_c40.md` 8cfff1c...; `data/v4/diagnostics/c34_por_delta_proof.json` 3b0e4d95...; Peach Dream stem manifest c4944ee8...

Four informational disclosures at Cycle 55 audit:

- **I-1 spec doc PRESENT on disk.** Positive downstream signal; endorsed as I-5-closed for c46+ M-V4-PROFILES readiness scoping.
- **I-2 worker-side D-3 misreading pattern.** Fifth consecutive cycle where the worker re-asserts a proposal-doc SHA cross-reference as a brief-side error when the brief itself cites the correct on-disk value; audit cross-check confirms the brief cites 8cffc1cecf8fed87... correctly. Consistent with the c44 audit's I-3 diagnosis; recommendation to break the recurrence chain either through an explicit brief flag or by the audit ceasing to re-note.
- **I-3 cosmetic v3-era cycle-45 naming collision.** Ten pre-existing ledger rows tagged cycle=45 at lines 727-736 from an earlier M-EAR-1 real-label-training v2 arc. No event_id UUID5 content-hash collision — different content, same milestone_id label class. v3→v4 campaign-continuity naming artifact, not a defect. Recommend next brief note this so future audit trails distinguish v3-era from v4-era cycle_45 rows via event content pinning.
- **I-4 closure-vs-preservation scope divergence.** Fourteenth consecutive cycle re-disclosing the standing divergence between the M-V4-CLOSE user-directive framing (END THE RUN cleanly, five novel songs) and the preservation-only scope of the current brief. Correct posture per FD-1 (brief is binding); six preserved escalations continue to gate closure work. Divergence is operator/researcher territory, not blocking.

Audit outcome

VALIDATED at range close across all nine validators (a)-(i). Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Four informational disclosures, none blocking.

The audit explicitly endorses three range-close positions: (a) the worker's honest closure of I-5 as PRESENT on disk (positive downstream signal); (b) the I-2 canonical P0 sidecar adoption executed correctly across the range; (c) the I-1 narrative counter realignment adopted at Cycle 55 so the brief and on-disk chain agree.

Discussion

Three things about this range are worth naming.

First, the range demonstrates that formalizing preservation surfaces as on-disk selection events strengthens auditability without changing the cadence's substance. Both the Priority 0 escalation-preservation sidecar and the Priority 5 Track B/C/D deferral-preservation rollout pattern were previously tracked in cycle narrative — the discipline was correct, but the audit surface was narrative rather than artifactual. Making them on-disk _selection/ events with string-typed supersedes_path chains gives the same audit surface every other preservation surface already has: a discoverable pre-vs-post byte-identity check across the six memos and the four deferral rows. The Priority 8 consolidation-proposal HOLD pattern extended the same treatment to the fourth operator-authority surface. By Cycle 55 close the range emits five string-typed supersede events plus one fresh-hold null-typed supersede event per cycle, all exercising the c14 lemma; the range is now hitting six different discipline surfaces on every cycle rather than only two.

Second, the I-5 discovery on the spec document is a rare positive downstream signal in a preservation-only cadence. The document had been named across the two prior briefs as an ABSENT scoping blocker for M-V4-PROFILES substantive advance; the Cycle 55 fresh probe found it PRESENT. This is not a change the campaign engineered — the worker did not author the doc, and doing so would have violated the FD-1 authoring boundary — but a change the campaign detected honestly and forwarded to the next research pass. The pattern shows that the preservation-only cadence's fresh-probe discipline on filesystem state (as opposed to just chain-superseding what was true last cycle) actually catches externally-driven state

changes when they occur. This matters because it means the campaign is not just holding — it is watching, and it will notice the state change that eventually unblocks it.

Third, the I-2 recurring worker-side misreading pattern is worth naming as a small but real audit cost. Five consecutive cycles have now had the worker re-assert the same proposal-doc SHA cross-reference as a brief-side error when the brief itself cites the correct value. Each individual instance is harmless — both anchors are byte-identical to the landing state, no drift is introduced — but the audit is now noting the same non-defect five times, which is a cost even under a zero-blocking-findings outcome. The audit's recommendation is either an explicit brief flag to break the chain or an audit-side agreement to stop re-noting. The larger discipline point is that when a preservation cadence stabilizes, workers can develop pattern-matched expectations that drift from what the brief actually says; the correct fix is either a brief-side signal (make the “on-disk is X; NOT Y” pattern more prominent) or a worker-side habit change (read the brief afresh each cycle rather than relying on memory of the pattern). Both are recoverable in one cycle if flagged explicitly.

Open questions

- **Composite-FP-drift operator adjudication.** Three named paths (A accept render-level bar via new invariant (f); B hold strict; C harden objective.py). Path A remains the single-action resolution across all three fine-fit drivers. No adjudication in `live_guidance` across the range.
- **Consolidation-proposal operator selection.** Three named options (OPT_1 / OPT_2 / OPT_3) in `docs/v4_por_consolidation_strategy_proposal_c40.md`. No selection token in `live_guidance` across the range.
- **Non-CG bass acceptance-policy escalation.** Remains `blocked_on_operator=true`. SF2_CONFIRMED FORBIDDEN. Systematic 4-arc composite-vs-source-of-truth finding still predicts OPT2 (refuse extension + OPT3 htdemucs bass fallback) as the invariant-compliant outcome.
- **Metric-semantics escalation.** Remains `blocked_on_operator=true` with `carried_from_cycle=16` at Cycle 55 close.
- **Emitter writer-boundary chain.** Eleven-cycle chain-supersede continues if `long_exposure/` remains ABSENT. Canonical M-1 naming holds across the range.
- **POR shadow-drift stand-pat.** Twelve-cycle chain continues absent operator-supplied c31/c32 POR snapshot. c34 empirical proof transitively holds.
- **Consolidation-proposal HOLD chain.** Established across two cycles this range (Cycle 54 first HOLD; Cycle 55 chain-supersede). Continues absent operator selection token.
- **M-V4-PROFILES readiness re-scoping (positive downstream signal).** `docs/specs/v4_sound_matching_layer_spec.md` now PRESENT on disk (I-5 closed). Next-cycle researcher may re-scope M-V4-PROFILES readiness if operator P3 adjudication also lands.
- **Worker-side D-3 misreading pattern.** Recommended remediation via explicit brief flag or audit-side agreement to stop re-noting.
- **v3-era vs v4-era cycle-45 naming collision.** Cosmetic; recommend next brief note the artifact so future audit trails distinguish via event content pinning.
- **Expected next-cycle open state.** `parseable_milestones = 915`; six escalation memo SHAs unchanged; consolidation proposal SHA unchanged at `8cffc1c...`; spec doc PRESENT on disk (I-5 closed); `long_exposure/` ABSENT; c45 sidecars on disk byte-identical; 48/48 test suite green pre-extension; chain-length P1 = 12, P2 = 13, P8 HOLD = 3 at next-cycle close; 50/50 test target; escalation counters {17, 17, 17, 16, 16, 15}.
- **Four operator triggers that would change the state.** (a) A c31/c32 POR snapshot for Priority 2 close; (b) a PATH_A / PATH_B / PATH_C adjudication token for the composite-FP-drift memo; (c) an OPT_1 / OPT_2 / OPT_3 selection token for the consolidation proposal; (d)

resolution of the non-CG bass acceptance policy. All four remain absent.

Appendix: Provenance

Directive. Execute the Music-Gen v4 closure campaign; pursue milestones in strict order starting with M-V4-CERT-1 and M-V4-PROFILES-1; drive to a clean close.

Cycle range. cycles 53-55.

Working directory. /home/user/long-exposure-runs/music-gen.

Session references.

- Cycle 53 researcher f3c5608b-eb92-4fcf-8828-65344bf20851; worker 13808a1d-6644-490a-8bc7-ed793c7e24ba; auditor df856c6d-9f5b-461f-a730-b4d25f6b803f.
- Cycle 54 researcher fec80391-4717-4c62-8510-9d0d82a4f25a; worker 3d6c1b5a-f883-4af0-8315-d9774620b81c; auditor 68018633-91b8-4095-a475-cbe26c4c8805.
- Cycle 55 researcher 595675c1-2299-4fe7-b4e8-538ec093db95; worker 3c411270-8b9e-4b73-ba59-a7ef21bb03b4; auditor 5df5094e-3e97-4d4d-bcf6-4b645cc4e15c.

Audit verdict. VALIDATED at range close across all nine validators (a)-(i). Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Four informational disclosures: I-1 spec doc PRESENT on disk (positive downstream signal; endorsed as I-5-closed); I-2 worker-side D-3 misreading pattern (fifth consecutive cycle recommendation to break the chain); I-3 cosmetic v3-era vs v4-era cycle-45 naming collision (no event_id UUID5 collision); I-4 standing closure-vs-preservation scope divergence carried across fourteen consecutive cycles.

Terminal deliverables landed this range.

- Three Priority 1 chain-supersede preservation events at data/v4/_selection/c{43,44,45}-emitter-writer-boundary-preservation.json per M-1 canonical naming, each string-typed supersedes_path → prior-cycle actual on-disk filename per c14 lemma.
- Three Priority 2 stand-pat preservation events at data/v4/_selection/c{43,44,45}-por-drift-preservation.json.
- Canonical Priority 0 escalation-preservation sidecar pattern established at Cycle 53 and chain-superseded across Cycles 54 and 55 (Cycle 55: c45-escalation-preservation.json chain-supersedes c44 sha 608e8138...).
- Canonical Priority 5 Track B/C/D deferral-preservation rollup sidecar pattern established at Cycle 53 and chain-superseded across the range (Cycle 55: c45-track-bcd-deferral-preservation.json sha db8bfed7...).
- Canonical Priority 8 consolidation-proposal HOLD pattern established at Cycle 54 (c44-consolidation-proposal-hold.json sha f70b8ab1...) and chain-superseded at Cycle 55 (c45-consolidation-proposal-hold.json).
- Three Priority 6 fresh-hold events with supersedes_path: null per c14 lemma.
- Twelve honest-deferral rows total across the range for Track B/C/D (four per cycle) with concrete resume commands.
- Six new test cases across the range (two per cycle) extending tests/test_c30_legacy_mode_regression.py in-place 34 → 36 → 38 → 40; standalone tests/test_fine_fit_serial_lock_c32.py unchanged at 8/8; cross-cycle total 48/48 PASS at range close.
- I-1 narrative counter realignment adopted at Cycle 55: {15, 15, 15, 14, 14, 13} → {16, 16, 16, 15, 15, 14}.
- I-5 spec-doc PRESENT-on-disk disclosure via c45-consolidation-proposal-hold.json.i5_binding_doc_absent_check.
- Three housekeeping-triad sequences (one per cycle): _run/cycle_N_closed + _archive/cycle-N-scratch/ + _infra/adopt-cycleN-tests; emitter-sentinel guards tools/.c{43,44,45}_ledger_

emitted.

Ledger + POR arithmetic across the range. Cycle 53 +14 rows; Cycle 54 +14; Cycle 55 +14. POR trajectory 873 → 887 → 901 → 915. Each cycle open matches prior cycle close (delta zero). POR baseline 745 maintained.

Read-only anchors preserved byte-identical pre-vs-post (9 verified at range close).

- scripts/sound_match/objective.py 8087ce80...
- scripts/sound_match/_sweep_hygiene_c27.py 771ff42b...
- scripts/sound_match/_serial_lock_op1.py 121809db...
- data/v4/deliveries/31a164f845f8e27e/cg_ab_mix.wav 6e13e007...
- docs/agent_picks_selection_invariants.md 29a1610b...
- docs/emitter_exemption_policy.md fd2c33a7...
- docs/v4_por_consolidation_strategy_proposal_c40.md 8cffc1c...
- data/v4/diagnostics/c34_por_delta_proof.json 3b0e4d95...
- Peach Dream stem manifest c4944ee8...

Chain-supersede lineage at range close.

- Priority 1 (emitter-writer-boundary): c34 fork → 11 consecutive chain-supersedes → c45.
- Priority 2 (POR shadow-drift stand-pat): c34 empirical proof → 12 consecutive stand-pats → c45.
- Priority 0 (escalation-preservation): c43 canonical adoption → 3 consecutive sidecars → c45.
- Priority 5 (Track B/C/D deferral rollup): c43 canonical adoption → 3 consecutive sidecars → c45.
- Priority 8 (consolidation-proposal HOLD): c44 canonical adoption → 2 consecutive HOLDs → c45.

Six operator escalations preserved verbatim with I-1 realigned carried_from_cycle values at Cycle 55 close: non-CG bass 16, metric-semantics 16, drums-fine 16, v2-bass-fine 15, guitar-fine 15, composite-FP-drift 14. All blocked_on_operator=true. _manager/ exhaustive at six.

Substantive-heartbeat streak at range close. c33 → c45 = 14 consecutive substantive-heartbeat cycles under c36 auditor terminal contract.

Environment pin. Canonical 7-key env_pin_sha256=2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842 unchanged this range; FD-16(a) re-issue not triggered. FD-16(c) replay-proof ×2 per render family per song invariant carried; no new render surfaces.

Discipline guards asserted (AST-scannable). No PRNG imports, no sidecar_nonfactor, no VST3 state APIs, no --verify-det bypass, /usr/bin/python3 interpreter guard implicit via c34+ pattern. supersedes_path typed as string throughout (except fresh-hold P6 with null per c14 lemma); six string-typed supersede events plus one null-typed supersede at Cycle 55 alone. Zero SF2_CONFIRMED verdicts on non-CG bass. OP-1 serial-lock invariant in force; 8/8 standalone lock tests continue to guard. No wait-on-operator memo (banned per operator directive 2026-09-03 part 2); Priority 0 preserve-only posture satisfies genuine operator-authority carve-out.

Milestone status at range close.

- M-V4-CERT-1 — validated (E2E_DETERMINISM_HOLDS on the v3 spine).
- M-V4-PROFILES-1 CG (5/5 instruments) — validated.
- M-V4-PROFILES-1 non-CG bass — 2/4 SF2_RULED_OUT (Rome, Peach Dream); 2/4 STILL_INDETERMINATE (WIG, Disco A). Stage-2 re-work gated on operator adjudication. docs/specs/v4_sound_matching_layer_spec.md PRESENT on disk (I-5 closed; positive readiness signal for next-cycle scoping if operator P3 adjudication also lands).

- M-V4-PROFILES-1 non-CG drums — 0/4 (Track D unblocked at driver-regression level; stage-2 gated on operator adjudication).
- M-V4-PROFILES-1 non-CG guitar — 0/2 (WIG + Peach Dream guitar are NULL by earlier MIDI-probe).
- M-V4-SHOWCASE-1 CG — LANDS_pending_operator (cg_ab_mix.wav SHA 6e13e007... byte-identical since c17).
- M-V4-SHOWCASE-1 non-CG — BLOCKED on non-CG bass acceptance-policy escalation.
- M-V4-RULES-1 — scaffold landed c20; substantive implementation queued.
- M-V4-EAR-1 — not yet opened.
- M-V4-GEN-1 — conditional on M-V4-RULES + M-V4-EAR.
- M-V4-CLOSE-1 — c24 amendment landed; further amendments as substantive work completes.

Next-cycle first task. Continue the nine-priority preservation cadence per c45 auditor forward guidance. Expected chain-lengths at next-cycle close: P1 = 12, P2 = 13, P0 = 4 sidecars, P5 = 4 sidecars, P8 HOLD = 3 sidecars. Expected 50/50 test target. Expected POR at next-cycle open = 915 (matches Cycle 55 close). Escalation counters bump to {17, 17, 17, 16, 16, 15}. Next-cycle researcher may re-scope M-V4-PROFILES readiness given I-5 closed if operator P3 adjudication also lands. If any operator adjudication token arrives via `live_guidance`, execute the chosen path. Recommend next-cycle brief either explicitly flag the I-2 D-3 pattern to break the recurrence chain, or the audit side stops re-noting. Recommend next-cycle brief note the cosmetic v3-era vs v4-era cycle-45 naming artifact so future audit trails distinguish via event content pinning. Operator ear remains LANDS authority post-hoc per FD-6.